ARC NCP: Adaptations for Anthropic

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The Alignment Research Center (ARC) hopes to explain the behaviour of trained neural networks by short, machine-checkable descriptions of their internal structure, even where no formal proof of that behaviour exists. As a test of whether such explanations should exist at all, Neyman [1] formulated a computational version of the principle that an outrageous coincidence in mathematics has a reason. Write $X_0 \subseteq \{0,1\}^{3n}$ for the 2^{2n} strings ending in n zeros. For a reversible circuit $C\colon \{0,1\}^{3n} \to \{0,1\}^{3n}$, let $P(C)$ be the property that $C(x) \notin X_0$ for every $x \in X_0$. We call this property the *coincidence*. Let U_n be the distribution of random reversible circuits with n^2 layers, in which each layer partitions the $3n$ wires uniformly at random into n triples and applies to each triple an independent bijection of $\{0,1\}^3$ chosen uniformly among all 8! [1, footnote 3].

Conjecture 1 (No-coincidence principle [1])**.** *There is a polynomial-time verifier V that takes as input a reversible circuit $C\colon \{0,1\}^{3n} \to \{0,1\}^{3n}$ and a string π, such that for every n:*

1. *(completeness) for every C with $P(C)$ there is a π of length polynomial in the size of C with $V(C,\pi) = 1$;*
2. *(soundness) for at least 99% of the circuits $C \sim U_n$, there is no π with $V(C,\pi) = 1$.*

If a circuit C behaved like a uniformly random permutation, each of the 2^{2n} inputs in X_0 would land in X_0 with probability 2^{-n}, so $P(C)$ would hold with probability about $(1-2^{-n})^{2^{2n}} \approx e^{-2^n}$. The set of circuits with the property P is coNP-complete: it is in coNP, since an input $x \in X_0$ with $C(x) \in X_0$ witnesses that $P(C)$ fails, and it is coNP-hard [1]. Soundness only asks the verifier to reject most random circuits, not every circuit without P. If it had to accept exactly the circuits with P, the set of these circuits would be in NP, so this stronger statement is equivalent to NP = coNP. In particular, NP = coNP implies Conjecture 1, and a refutation of Conjecture 1 would prove NP $\neq$ coNP. Closely related *heuristic proof systems* were studied by Itsykson and Sokolov [2].

Andrea Lincoln proposed the following analogue of Conjecture 1 for Erdős–Rényi graphs (unpublished). For a constant c, let $k = k(n) = \lceil c \log_2 n \rceil$, and let $Q_k(G)$ be the property that the graph G has no clique of size k; this is the coincidence for graphs. The clique number of $G(n,1/2)$ is $(2-o(1))\log_2 n$ with high probability [3, 4], so for $c < 2$ the property Q_k is rare.

Conjecture 2 (Clique NCP)**.** *Let $1 < c < 2$. There is a polynomial-time verifier V such that for every n:*

1. *(completeness) for every graph G on n vertices with $Q_k(G)$ there is a π of length polynomial in n with $V(G,\pi) = 1$;*
2. *(soundness) for at least 99% of the graphs $G \sim G(n,1/2)$, there is no π with $V(G,\pi) = 1$.*

Why $1 < c < 2$. For $c < 1$, the greedy algorithm finds a clique of size $(1 - o(1)) \log_2 n$ with high probability [5], so a verifier can run it and accept if it fails, with no certificate. For $1 < c < 2$, cliques of size k exist with high probability, but no polynomial-time algorithm is known that finds a clique of size $(1 + \delta) \log_2 n$ for any constant $\delta > 0$. Like P, the property Q_k is in coNP, with a k-clique as the witness that it fails, so a refutation of Conjecture 2 would also prove NP $\neq$ coNP. Unlike P, it can be decided in time $n^{O(\log n)}$ by exhaustive search, so it is not coNP-hard unless coNP has quasi-polynomial-time algorithms.

Main questions

Question 1. *Prove or refute Conjecture 1. A refutation may assume any standard or otherwise well-motivated complexity-theoretic, cryptographic or average-case hardness assumption.*

Question 2. *Prove or refute Conjecture 2, for some or all $c \in (1, 2)$, again possibly under a hardness assumption.*

Since a refutation of either conjecture would prove NP $\neq$ coNP, an unconditional refutation is out of reach, and a refutation from an assumption is the natural goal.

Background on the Clique NCP

We record three observations on Conjecture 2: a natural certificate that is not complete, a relaxation that is too easy, and an approach that runs into a known barrier.

Colourings. A proper colouring of G with $k - 1$ colours certifies $Q_k(G)$, because a clique has at most one vertex of each colour. The verifier that accepts such colourings is sound, since $G(n, 1/2)$ has chromatic number $(1 + o(1))\, n/(2 \log_2 n)$ with high probability [6]. It is not complete: there are graphs with no triangle and arbitrarily large chromatic number [7], so some graphs with Q_k are not $(k - 1)$-colourable.

False negatives. Suppose completeness is weakened so that the verifier may reject a small fraction of the graphs with Q_k. Then the following verifier, which uses no certificate, appears to suffice: accept G if it has at least $2n$ fewer edges than $\binom{n}{2}/2$. By Hoeffding's inequality, at most an e^{-16} fraction of all graphs is accepted. Conditioning on Q_k should lower the number of edges by far more than $2n$, so one expects that only a vanishing fraction of the graphs with Q_k is rejected; we have not checked a proof.

Hypergraph containers. Let $\mathcal{H}$ be the hypergraph whose vertices are the $N = \binom{n}{2}$ pairs of vertices of K_n, and whose hyperedges are the edge sets of the k-cliques of K_n. A graph G has the property Q_k exactly when its edge set is an independent set of $\mathcal{H}$. Container lemmas [8, 9] state that, if the hyperedges of a uniform hypergraph are spread out well enough, there is a small family $\mathcal{C}$ of vertex sets, the containers, with three properties: every independent set lies in some container; every container contains few hyperedges; and the container of an independent set I is determined by a small subset of I, its fingerprint. This suggests a verifier for Conjecture 2 whose certificate is a fingerprint $T \subseteq E(G)$; the verifier computes the container $C(T)$ and accepts if $E(G) \subseteq C(T)$. Completeness follows from the first and third property, provided the container can be computed

from the fingerprint in polynomial time, as in the algorithmic proofs of the lemmas. For soundness, a union bound gives

$$\Pr_{G\sim G(n,1/2)}\big[\exists\, C\in\mathcal{C} : E(G)\subseteq C\big] \le |\mathcal{C}|\cdot\max_{C\in\mathcal{C}} 2^{-(N-|C|)},$$

A container with few k-cliques must miss many of the N pairs (a supersaturation statement), so $N-|C|$ is large; soundness needs $\log_2|\mathcal{C}|$ to be smaller than $N-|C|$ by a constant margin, for every container C. The obstacle is that $\mathcal{H}$ has uniformity $\binom{k}{2}=\Theta(\log^2 n)$. In the known container lemmas, the logarithm of the number of containers grows quadratically in the uniformity, and Campos and Samotij [10] identify this as a natural barrier for the second-moment type arguments behind these lemmas. Accordingly, the best known application to cliques, which shows that almost all K_{r+1}-free graphs are r-partite, works only for r much smaller than $\log n$ [11]. Applying containers for $k=c\log_2 n$ seems to need new ideas specific to cliques.

Further questions

The following questions are partial steps towards Questions 1 and 2.

How rare are the coincidences?

Completeness and soundness together force the coincidence to have probability at most 1% for every n, so each conjecture is false unless its coincidence is rare. For both coincidences, the naive heuristic that treats the relevant events as independent is far from the truth, because structured objects have the property for a visible reason and are much more likely than the heuristic predicts.

Circuits. Fix one of the last n wires. In each layer exactly one gate acts on it, and this gate is a uniformly random bijection of $\{0,1\}^3$. With probability $(4!)^2/8! = 1/70$, the gate maps the four inputs in which this wire is 0 onto the four outputs in which it is 1; with the same probability, it maps the four inputs in which the wire is 1 onto the four outputs in which it is 1. Suppose the gate in the first layer does the former and the gates in all later layers do the latter. On every input from X_0 the wire starts as 0, so it ends as 1, and C has the property P. The n^2 layers are independent, so

$$p_n := \Pr_{C\sim U_n}[P(C)] \ge 70^{-n^2} \approx 2^{-6.13\,n^2}.$$

No upper bound on p_n is known to us, not even $p_n \le 1\%$.

Cliques. Let $1<c<2$ and $q_n := \Pr_{G\sim G(n,1/2)}[Q_k(G)]$. The expected number of k-cliques in $G(n,1/2)$ is $\binom{n}{k}2^{-\binom{k}{2}} = n^{\Theta(\log n)}$, so treating the k-cliques as independent would suggest $q_n \approx \exp(-n^{\Theta(\log n)})$. Again structured graphs are far more likely: every $(k-1)$-partite graph has the property Q_k, and the graphs that are $(k-1)$-partite with respect to one fixed balanced partition already have probability $2^{-(1+o(1))\,n^2/(2(k-1))}$. On the other hand, Janson's inequality [12, Chapter 8] gives an upper bound. Hence

$$2^{-(1+o(1))\,n^2/(2(k-1))} \;\le\; q_n \;\le\; \exp\big(-\Omega(n^2/\log^4 n)\big),$$

so $\log(1/q_n)$ is known only up to a factor of order $\log^3 n$. For $k-1=r$ much smaller than $\log n$, almost all K_{r+1}-free graphs on n vertices are r-partite [13, 11], which determines the probability of being K_{r+1}-free up to a factor $2^{o(n^2/r)}$; our $k=c\log_2 n$ lies outside this range.

Question 3. *1. Prove upper and lower bounds on p_n that are as close as possible. Is $p_n\to 0$? Is $p_n \le 2^{-\Omega(n)}$? Is $p_n = 2^{-\Theta(n^2)}$, and if so, what is $\lim_{n\to\infty} -\frac{1}{n^2}\log_2 p_n$?*

2. *For $1 < c < 2$, determine $\log(1/q_n)$ up to a constant factor, and ideally up to a factor $1+o(1)$. Is $q_n = 2^{-(1+o(1))\,n^2/(2(k-1))}$, that is, are $(k-1)$-partite graphs the dominant reason for having no k-clique, as in [13] for smaller cliques?*

3. *In both cases, which structured objects dominate? Is there an explicit NP set S_n of objects with the coincidence (such as circuits with a stuck wire, or graphs given together with a proper $(k-1)$-colouring) whose probability is $p_n^{1+o(1)}$, respectively $q_n^{1+o(1)}$?*

Weaker soundness, and how much 99% matters

For $\varepsilon = \varepsilon(n) \in (0,1)$, write $\mathrm{NCP}(\varepsilon)$ for Conjecture 1 with soundness "for at least a $1-\varepsilon$ fraction of the circuits $C \sim U_n$, there is no π with $V(C,\pi) = 1$", so that Conjecture 1 is $\mathrm{NCP}(1/100)$, and define $\mathrm{Clique}(\varepsilon)$ from Conjecture 2 in the same way.

A proof of $\mathrm{NCP}(\varepsilon)$ for some ε much larger than $1/100$ would already be a good partial answer to Question 1. Even this is not trivial: the verifier has to reject a non-negligible fraction of random circuits with certainty, while accepting every circuit with P. A verifier that rejects only when it finds, among polynomially many inputs $x \in X_0$, one with $C(x) \in X_0$, rejects only about a $\mathrm{poly}(n) \cdot 2^{-n}$ fraction of random circuits, if these behave like random permutations.

In the other direction, De Bona [14] showed that $\mathrm{NCP}(1/100)$ implies $\mathrm{NCP}(\varepsilon)$ for every fixed $\varepsilon > 0$ under an additional assumption on polynomial-time reductions to the set of circuits without P;[1] the running time of the amplified verifier grows exponentially in $1/\varepsilon$ unless the reductions have degree 1.

Question 4. 1. *For which $\varepsilon < 1$ can $\mathrm{NCP}(\varepsilon)$ be proved, unconditionally or from a standard assumption? In particular, does $\mathrm{NCP}(1 - 1/\mathrm{poly}(n))$ hold, or $\mathrm{NCP}(1/2)$? The goal is a verifier with soundness as large as possible; at $\varepsilon = 1/100$ this is Question 1 itself. The same questions for $\mathrm{Clique}(\varepsilon)$.*

2. *Does $\mathrm{NCP}(1/100)$ imply $\mathrm{NCP}(\varepsilon)$ for every constant $\varepsilon > 0$, without additional assumptions?*

3. *For which functions $\varepsilon(n) \to 0$ does $\mathrm{NCP}(\varepsilon(n))$ follow from Conjecture 1, or from a standard complexity-theoretic assumption together with Conjecture 1?*

4. *Is there a sequence of distributions D_n on reversible circuits with $\Pr_{D_n}[P] \to 0$ for which the statements $\mathrm{NCP}(\varepsilon)$, defined with D_n in place of U_n, are not all equivalent for constant $\varepsilon \in (0,1)$, conditional on a standard assumption such as $\mathsf{coNP} \not\subseteq \mathsf{NP}/\mathrm{poly}$?*

Towards a refutation

Question 5. 1. Assumptions. *Under which hypotheses do the conjectures fail? Candidates are worst-case (an infinite polynomial hierarchy, $\mathsf{coNP} \not\subseteq \mathsf{NP}/\mathrm{poly}$), cryptographic (one-way functions, pseudorandom permutations, indistinguishability obfuscation) or average-case (hardness of finding or certifying cliques in $G(n,1/2)$). Conversely, for such a hypothesis H, is there an oracle relative to which H and Conjecture 1 both hold, with circuits allowed to contain oracle gates? Such an oracle rules out relativizing proofs that H refutes Conjecture 1.*

[1] De Bona calls it *reduction-regularity*. Roughly, it asks for a fixed $\delta > 0$ such that every NP set containing at least 99% of the circuits without P has a polynomial-time reduction to the set of circuits without P whose images keep at least a δ fraction of their probability under U_n; see [14] for the precise statement.

2. Hiding a coNP fact in a random-looking circuit. *Is there a randomised polynomial-time map f from Boolean formulas to circuits such that $f(\varphi, r)$ has P for every unsatisfiable φ, and for satisfiable φ the distribution of $f(\varphi, \cdot)$ gives every circuit at most $\gamma < 100$ times its probability under U_n? Together with Conjecture 1 such a map would put unsatisfiability in* AM *(Arthur draws r, Merlin sends a certificate for $f(\varphi, r)$), so the polynomial hierarchy would collapse [15].*

3. Incompressible circuits. *If every circuit with P has a short description, the conjecture holds: encode circuits so that U_n is uniform on 2^m codes (m need not be an integer), and suppose every circuit with P is output by a program of length at most $m - t$ that runs within a fixed polynomial time bound. Then the outputs of these programs form an* NP *set of probability less than 2^{-t+1} that contains all circuits with P, so Conjecture 1 holds if $t \geq 8$. Conversely, does* $\mathrm{NCP}(2^{-t})$ *imply that every circuit with P has polynomial-time nondeterministic distinguishing complexity*[2] *at most $m - t + o(t)$? If so, a family of circuits with P whose distinguishing complexity is at least $m - o(t)$ would refute* $\mathrm{NCP}(2^{-t})$*. Does such a family exist, for some $t(n) \leq 6.13\, n^2$?*

4. Restricted verifiers. *Fix a polynomial-size CNF formula F_C that is unsatisfiable exactly when $P(C)$ holds. Is there a family of circuits with P for which F_C has no polynomial-size resolution refutation? More ambitiously, show that for every polynomial-time test T with $\Pr_{C \sim U_n}[T(C) = 1] \leq 1\%$, the verifier that accepts if $T(C) = 1$ or if π is a polynomial-size resolution refutation of F_C is not complete. The same questions for other proof systems, such as cutting planes or sums of squares, and for Q_k.*

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[2] The length of a shortest program that, run nondeterministically within a given time bound, accepts the given string and no other string [16]. Language compression [17] bounds it for members of small sets that the program can query.

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